AER1319: FINITE VOLUME METHODS FOR CFD

Assignment #2

For Assignment #2, you are to implement and test the exact solution procedure for the Riemann initial value problem (IVP) of one-dimensional form of the Euler equations as discussed in class, i.e., the method of Gottlieb and Groth [1988]. The Riemann initial value problem (IVP) is defined as: given the one-dimensional (1D) form of the conservation equations

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{0}\,,$$

defined on the (x,t) domain $-\infty < x < \infty$ and $t > 0$ for position coordinate, x, and time, t, with discontinous initial data of the form

$$\mathbf{U}(x, t=0) = \mathbf{U}_0(x) = \begin{cases} \mathbf{U}_\mathrm{l} & \text{for } x \le 0\,, \\ \mathbf{U}_\mathrm{r} & \text{for } x > 0\,, \end{cases}$$

where the vectors of conserved solution variables, $\mathbf{U}$ and solution fluxes, $\mathbf{F}$, are given by

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \dfrac{p}{\gamma - 1} + \dfrac{\rho}{2} u^2 \end{bmatrix}, \qquad \mathbf{F} = \begin{bmatrix} \rho u \\ p + \rho u^2 \\ \rho u \left(\dfrac{\gamma p}{\rho(\gamma - 1)} + \dfrac{u^2}{2} \right) \end{bmatrix},$$

and where ρ, u, p, and γ are the density, velocity, pressure, and specific heat ratio, respectively, the self-similar solution $\mathbf{U}(x,t)$ is sought for $t > 0$.

To help you with this assignment, the C++ code:

`euler1d-riemann.cc` (download from course web site)

is provided which contains an implementation of the exact Riemann solver of Gottlieb and Groth [1988] (the paper by Gottlieb and Groth is also available for download from the course web site, see file `JCP-1988-gottlieb.pdf`). This method must be re-implemented using either Matlab or C, C++ (my graduate students must use either C or C++), as the case may be. The accurate performance of your implementation of the Riemann solver should be demonstrated for a number of Rieman IVPs through comparisons to results obtained using the C++ code provided. You should provide comparisons of the computed solutions for several Riemann problems corresponding to each of the four possible wave patterns (RCR, SCR, RCS, and SCS).

Please submit the code for the Riemann solver along with your results when you turn in your assignment.